12.60

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

A square matrix $\mathbf{B}$ is skew-symmetric if

a) $\mathbf{B}^\top = -\mathbf{B}$ b) $\mathbf{B}^\top = \mathbf{B}$ c) $\mathbf{B}^{-1} = \mathbf{B}$ d) $\mathbf{B}^{-1} = \mathbf{B}^\top$

Solution:

$\mathbf{B}$ is skew-symmetric, if and only if

$$\mathbf{B}^\top = -\mathbf{B} \tag{1}$$

Example:

$$\mathbf{B} = \begin{pmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{pmatrix} \quad \text{is a skew symmetric matrix} \tag{2}$$

$$\implies \mathbf{B}^\top = \begin{pmatrix} 0 & -2 & 1 \\ 2 & 0 & -3 \\ -1 & 3 & 0 \end{pmatrix} = -\mathbf{B} \tag{3}$$